Two almost planetary mass survivors of common envelope evolution

S. G. Parsons,[1★] A. J. Brown,[2] S. L. Casewell,[3] S. P. Littlefair,[1] J. van Roestel,[4] A. Rebassa-Mansergas,[5,6] R. Murillo-Ojeda,[7] M. A. Hollands,[8] M. Zorotovic,[9] N. Castro Segura,[8] V. S. Dhillon,[1,10] M. J. Dyer,[1] J. A. Garbutt,[1] M. J. Green,[11] D. Jarvis,[1] M. R. Kennedy,[12] P. Kerry,[1] J. McCormac,[8] J. Munday,[8] I. Pelisoli,[8] E. Pike[1] and D. I. Sahman[1]

[1]*Astrophysics Research Cluster, School of Mathematical and Physical Sciences, University of Sheffield, Sheffield S3 7RH, UK*
[2] *Hamburger Sternwarte, University of Hamburg, Gojenbergsweg 112, 21029 Hamburg, Germany*
[3]*School of Physics and Astronomy, University of Leicester, University Road, Leicester LE1 7RH, UK*
[4]*Anton Pannekoek Institute for Astronomy, University of Amsterdam, 1090 GE, Amsterdam, The Netherlands*
[5]*Departament de Física, Universitat Politècnica de Catalunya, c/Esteve Terrades 5, 08860 Castelldefels, Spain*
[6]*Institut d'Estudis Espacials de Catalunya, c/ Esteve Terradas, 1, Edifici RDIT, Campus PMT-UPC, 08860 Castelldefels, Spain*
[7]*Centro de Astrobiología (CAB), CSIC-INTA, Camino Bajo del Castillo s/n, campus ESAC, 28692 Villanueva de la Cañada, Madrid, Spain.*
[8]*Department of Physics, University of Warwick, Gibbet Hill Road, Coventry, CV4 7AL, UK*
[9]*Instituto de Física y Astronomía, Universidad de Valparaíso, Av. Gran Bretaña 1111, Valparaíso, Chile*
[10]*Instituto de Astrofisica de Canarias, E38205 La Laguna, Tenerife, Spain*
[11]*Max-Planck-Institut für Astronomie, Königstuhl 17, D-69117 Heidelberg, Germany*
[12]*School of Physics, University College Cork, Cork, T12 K8AF, Ireland*

arXiv:2501.14333v1 [astro-ph.SR] 24 Jan 2025

Accepted 2025 January 23. Received 2025 January 23; in original form 2024 December 17

ABSTRACT

White dwarfs are often found in close binaries with stellar or even substellar companions. It is generally thought that these compact binaries form via common envelope evolution, triggered by the progenitor of the white dwarf expanding after it evolved off the main-sequence and engulfing its companion. To date, a handful of white dwarfs in compact binaries with substellar companions have been found, typically with masses greater than around 50 M_{Jup}. Here we report the discovery of two eclipsing white dwarf plus brown dwarf binaries containing very low mass brown dwarfs. ZTF J1828+2308 consists of a hot (15900 ± 75 K) 0.610 ± 0.004 $M_\odot$ white dwarf in a 2.7 hour binary with a 0.0186 ± 0.0008 $M_\odot$ (19.5 ± 0.8 M_{Jup}) brown dwarf. ZTF J1230−2655 contains a cool (10000 ± 110 K) 0.65 ± 0.02 $M_\odot$ white dwarf in a 5.7 hour binary with a companion that has a mass of less than 0.0211 $M_\odot$ (22.1 M_{Jup}). While the brown dwarf in ZTF J1828+2308 has a radius consistent with its mass and age, ZTF J1230−2655 contains a roughly 20 per cent overinflated brown dwarf for its age. We are only able to reconstruct the common envelope phase for either system if it occurred after the first thermal pulse, when the white dwarf progenitor had already lost a significant fraction of its original mass. This is true even for very high common envelope ejection efficiencies ($\alpha_{CE} \sim 1$), unless both systems have extremely low metallicities. It may be that the lowest mass companions can only survive a common envelope phase if it occurs at this very late stage.

Key words: binaries: eclipsing – stars: fundamental parameters – white dwarfs – brown dwarfs

1 INTRODUCTION

Present day white dwarfs are the descendants of mostly A and F type stars, of which around 50 per cent are found in binary systems (e.g. De Rosa et al. 2014). It is therefore unsurprising that a significant fraction of white dwarfs are found in binaries (Toonen et al. 2017). In many of these systems the separation between the white dwarf and its companion is smaller than the size of the white dwarf progenitor when it was a giant, leading to the idea that these binaries must have originally been wider and they were subsequently brought closer as a result of the evolution of the white dwarf progenitor. It is generally accepted that this orbital shrinkage is usually the result of a common envelope (CE) phase (Paczynski 1976; Webbink 1984; Ivanova et al. 2013), caused by the giant star filling its Roche lobe and transferring mass to its companion. If the companion star cannot adjust quickly enough to this additional material then it too will fill its Roche lobe, resulting in an envelope of material surrounding the binary. This envelope will not necessarily be co-rotating with the binary and hence the stars experience a frictional force that extracts orbital energy from the system, causing the stars to spiral in towards each other. If sufficient orbital energy is transferred into the envelope then it can be ejected, leaving behind the now closer binary. If there is insufficient orbital energy, or energy is transferred from the orbit to the envelope very inefficiently, then the binary may instead merge during this phase.

Surviving the CE phase can be particularly challenging for substellar companions, such as brown dwarfs and planetary mass objects. Their lower masses mean they have less orbital energy available to

★ E-mail: s.g.parsons@sheffield.ac.uk

© 2025 The Authors

eject the envelope. Nevertheless, there are a small number of close white dwarf plus brown dwarf binaries known (11 at present, see French et al. 2024 for an up to date list), with around 0.1-0.5 per cent of white dwarfs thought to have brown dwarf companions (Farihi et al. 2005; Steele et al. 2011; Rebassa-Mansergas et al. 2019). This low number may reflect the difficulty in surviving the CE phase, but is primarily a result of the intrinsic rarity of brown dwarfs in binaries with main sequence stars with separations smaller than a few AU (i.e. the progenitors of close white dwarf plus brown dwarf binaries), the so-called "brown dwarf desert" (Grether & Lineweaver 2006; Grieves et al. 2017).

Brown dwarfs that do survive the CE phase and go on to closely orbit white dwarfs are extremely useful probes of binary evolution (Zorotovic & Schreiber 2022). When angular momentum loss through gravitational radiation brings these systems close enough for the brown dwarf to fill its Roche lobe, the system will form a cataclysmic variable (CV, Warner 1995; Knigge et al. 2011). These binaries therefore directly create systems that resemble period-bounce CVs, the end state of CVs that began mass transfer with more massive donor stars.

Brown dwarfs in close (orbital periods of less than a few days) binaries with white dwarfs can also be highly irradiated by the white dwarf, which can have effective temperatures 10-100 times larger than the brown dwarf. Since the brown dwarfs are also tidally locked, there are significant temperature variations between the hemisphere facing towards the white dwarf (the 'day' side) and the hemisphere facing away from white dwarf (the 'night' side). Temperature differences of hundreds of Kelvin are possible (Casewell et al. 2015; Hallakoun et al. 2023), making these objects extreme analogues of hot Jupiters. Given the difficulty in directly characterising the atmospheres of hot Jupiters, which are significantly outshone by their host stars at all wavelengths, the atmospheres of brown dwarfs irradiated by white dwarfs can be directly probed at infrared wavelengths, where they can outshine the white dwarf (e.g. Casewell et al. 2015; Longstaff et al. 2017; Lew et al. 2022; French et al. 2024), providing some insight into the otherwise-inaccessible atmospheric physics of hot Jupiter planets.

To date, the majority of close white dwarf plus brown dwarf binaries have been discovered by identifying apparently single white dwarfs with infrared excesses (e.g. Farihi et al. 2005; Steele et al. 2011; Girven et al. 2011). Given their small separations, these binaries are often found to be eclipsing, allowing precise measurements of the stellar and binary parameters (e.g. Littlefair et al. 2014). These studies typically find brown dwarf masses above around 50 M_{Jup}, potentially indicating that this is the lower limit for the mass of an object that can survive a CE phase (i.e. avoid merging with the core of the giant star). However, the recent discovery of the transiting planet candidate around the white dwarf WD 1856+534 in a 1.4 day period (Vanderburg et al. 2020) demonstrates that lower mass objects (<14 M_{Jup} for WD 1856+534b) can be found in close orbits around white dwarfs. While it is likely that this planet was scattered into its close orbit (Vanderburg et al. 2020), it is also possible that it survived a CE phase (Lagos et al. 2021), with the previous lower limit of around 50 M_{Jup} being a selection effect caused by the requirement for a detectable infrared excess. Lower mass (hence fainter and cooler) brown dwarfs and planetary mass objects can be completely outshone by white dwarfs even at infrared wavelengths and may only be detected if they happen to eclipse the white dwarf (or possibly via high precision radial velocity monitoring of apparently single white dwarfs).

The number of eclipsing white dwarf binaries has rapidly expanded in the last few years, thanks mainly to large scale, wide-field time-domain photometric sky surveys, such as the Zwicky Transient Facility (ZTF, Bellm et al. 2019; Graham et al. 2019; Masci et al. 2019), with hundreds of eclipsing systems now known (e.g. van Roestel 2019; Keller et al. 2022; Kosakowski et al. 2022; Brown et al. 2023). A small number of these eclipsing white dwarf systems sit on the white dwarf cooling track in the Gaia colour-magnitude diagram, implying that the companion to the white dwarf in these systems contributes a negligible amount of optical light compared to the white dwarf, making these excellent close white dwarf plus brown dwarf binary candidates.

In this paper we present follow up phase-resolved spectroscopy and high-speed photometry of two candidate white dwarf plus brown dwarf binaries. These were initially identified as white dwarfs with deep eclipses in ZTF and lacked any detectable excess flux at longer wavelengths, implying low mass or even substellar companions. Here we measure their stellar and binary parameters and find that both systems contain brown dwarfs with masses far below what is typically found in post-CE systems and so we test if standard CE reconstruction methods are able to reproduce these systems.

2 OBSERVATIONS AND THEIR REDUCTION

In this section we summarise our spectroscopic and photometric observations and their reduction. A full list of our observations is given in Table 1.

2.1 X-Shooter spectroscopy

We obtained phase-resolved medium resolution spectroscopy of our targets with the echelle spectrograph X-shooter (Vernet et al. 2011), which is mounted at the Cassegrain focus of the VLT-UT2 at Paranal, Chile, on the nights of 2024 April 10-12. X-shooter covers the spectral range from the atmospheric cutoff in the UV to the near-infrared K band in three separate arms, known as the UVB (0.30−0.56 microns), VIS (0.56−1.01 microns) and NIR (1.01−2.40 microns). Separate slit widths can be set for each arm and our observations were performed with slit widths of 1.0, 0.9 and 0.9 arcsec in the UVB, VIS and NIR arms respectively. We also binned the detector in the VIS arm by a factor of 2 in the spatial direction, while the UVB arm was binned by a factor of 2 in both the spatial and dispersion directions. This results in a resolution of R≃5000 in the UVB and R≃8000 in the VIS arm. In order to minimise slit losses we ran the active flexure compensation procedure once every hour and realigned the slit to the parallactic angle while doing so.

All of the data were reduced using the standard X-shooter pipeline release (version 3.6.3) within esoreflex (Freudling et al. 2013). Due to the faintness of our targets in the NIR arm (both are too faint at these wavelengths to be detected in the 2MASS survey) and the fact that we did not nod the telescope between observations, the data for both of our targets in the NIR are extremely poor and we do not include these data in any of our analysis.

The standard X-shooter pipeline can reach a velocity accuracy of around 8 km s^{-1} in the VIS arm. We improved this accuracy by fitting a number of telluric absorption features and corrected for small (typically ∼1 km s^{-1}) systemic velocity offsets in the data via the method described in Parsons et al. (2017a). This allowed us to reach an accuracy of a few km s^{-1} around the Hα line. These corrections predominantly affect our measurement of the systemic velocity of our systems, since accurate wavelength calibration is essential for this. Differences in the accuracy of the wavelength calibration from spectrum to spectrum are minimal, meaning that our radial velocity semi-

Table 1. Journal of observations.

Telescope/ Instrument	Date	Filters	No. of exposures	Exposure times (s)	Transparency	seeing (arcsec)
ZTF J1230−2655:						
VLT/X-shooter	2024-04-10	UVB/VIS/NIR	7/6/31	510/600/120	Clear	0.7–1.0
VLT/X-shooter	2024-04-11	UVB/VIS/NIR	20/18/92	510/600/120	Clear	0.5–0.7
VLT/X-shooter	2024-04-12	UVB/VIS/NIR	7/6/31	510/600/120	Thin clouds	1.0–1.2
NTT/ULTRACAM	2024-05-02	u_s,g_s,i_s	265/796/796	13.5/4.5/4.5	Clear	1.2–1.5
ZTF J1828+2308:						
VLT/X-shooter	2024-04-10	UVB/VIS/NIR	10/9/46	510/600/120	Clear	0.7–1.0
VLT/X-shooter	2024-04-11	UVB/VIS/NIR	10/9/46	510/600/120	Clear	0.5–0.7
GTC/HiPERCAM	2024-04-30	u_s,g_s,r_s,i_s,z_s	565/1695/1695/847/847	6/2/2/4/4	Clear	0.7–1.0

amplitude measurements can be more precise than this (<1 km s^{-1}). All spectra were then placed on a barycentric wavelength scale.

2.2 ULTRACAM and HiPERCAM photometry

ZTF J123016.59−265551.34 (listed as GALEX J123016.6−265551 in SIMBAD, hereafter ZTF J1230−2655) was observed with the high-speed camera ULTRACAM (Dhillon et al. 2007), mounted on the 3.5-m New Technology Telescope at La Silla, Chile on the night of 2024 May 2. ULTRACAM uses a triple beam setup and three frame transfer CCD cameras, which allows simultaneous data in three different wavebands with negligible (24 ms) dead-time between exposures. For our observations we made use of the high throughput super-SDSS u_s, g_s and i_s filters and targeted the orbital phases around the eclipse of the white dwarf. We used exposure times of 4.5 seconds in the g_s and r_s bands and longer exposure times of 13.5 in the u_s band.

We observed ZTF J182848.77+230838.0 (hereafter ZTF J1828+2308) with the high-speed quintuple band camera HiPERCAM (Dhillon et al. 2021) mounted on the 10.4-m Gran Telescopio Canarías (GTC) on La Palma, Spain on 2024 April 30. HiPERCAM allows simultaneous observations in the super-SDSS u_s,g_s,r_s,i_s and z_s bands with negligible (8 ms) dead-time between exposures. We used exposure times of 2 seconds in the g_s and r_s bands, 4 seconds in the i_s and z_s bands and 6 seconds in the u_s band.

Both the ULTRACAM and HiPERCAM data were reduced using the HiPERCAM pipeline software[1]. After bias subtraction and flat fielding (using twilight flats) the source flux was determined with aperture photometry using a variable aperture scaled according to the full width at half maximum (FWHM). The HiPERCAM z_s band data were also defringed using a reference fringe map. Transmission variations were accounted for by determining the flux relative to a comparison star in the field of view. The data were then flux calibrated using observations of the standard stars WD 1031−114 (for the ULTRACAM observations of ZTF J1230−2655) and Feige 66 (for the HiPERCAM observations of ZTF J1828+2308), using the reference magnitudes for the super-SDSS filters from Brown et al. (2022).

All times were converted to a TDB (Barycentric Dynamical Time) timescale corrected to the barycentre of the solar system. Using modified Julian dates format, this results in timing measurements of the form BMJD(TDB).

As noted by Brown et al. (2023), there is a faint source 2.8″ away from ZTF J1828+2308. Fortunately due to the good seeing during our HiPERCAM observations (~0.8″) the target and this source are well resolved and we masked out any pixels around this source to ensure that the final light curve and flux calibration were not affected by it.

[1] https://github.com/HiPERCAM/hipercam

3 RESULTS

3.1 ZTF J182848.77+230838.0

ZTF J1828+2308 was first listed as a candidate white dwarf plus brown dwarf binary by Kosakowski et al. (2022), who presented the ZTF light curve and noted the very deep eclipse of the white dwarf every 2.7 hours, as well as the lack of any clear excess in the PanSTARRS y band, indicating a likely substellar companion. Follow up high-speed multi-band ULTRACAM photometry of this object was presented by Brown et al. (2023), who used various theoretical relationships to constrain the mass of the white dwarf companion from the light curves, concluding that it was very likely substellar.

Our HiPERCAM light curves achieve a much greater depth than the ULTRACAM data in Brown et al. (2023), but still fail to detect the companion in any bands. By combining all observations taken during the eclipse, we place limits on the companion's brightness in the redder bands (where we might expect to detect a brown dwarf) of $i_s > 24.5$ and $z_s > 24.1$. Based on the Gaia DR3 parallax (4.91 ± 0.10 mas, Gaia Collaboration et al. 2023), this corresponds to absolute magnitude limits of $M_i > 17.9$ and $M_z > 17.6$, implying a spectral type later than L4 (Kiman et al. 2019).

While we do not detect the underlying photosphere of the companion to the white dwarf, we do detect Hα emission from its heated face. This, along with the radial velocity variations of the white dwarf itself, allow us to place strong constraints on the stellar masses, which combined with analysis of the white dwarf spectrum and eclipse light curves yield the full system parameters.

3.1.1 *Radial velocities*

ZTF J1828+2308 contains a fairly hot ($T_{\rm eff}$ = 15900 K) white dwarf. This, combined with the short orbital period of 2.7 hours (hence small orbital separation), means that the companion is moderately irradiated. The effect is fairly minor and there is not a clear reflection effect in the ZTF light curves or follow up high-speed data (although these are focused around the white dwarf eclipse phases, where irradiation effects are minimal). However, inspection of the X-shooter spectra clearly revealed Hα emission moving at high velocities and in anti-phase with the white dwarf absorption line (see Figure 1).